

## Response to Reviewers

**Paper ID 93:** Bellman-Guided Warm-Start Search for LSTM Configuration in ECG Forecasting

We thank the reviewers for identifying weaknesses in causal attribution, validation design, uncertainty reporting, scalability, data provenance, and related-work coverage. We revised the manuscript and reran a controlled five-seed experiment. Complete development traces are split before scaling/windowing; the official ECG5000 test partition remains untouched until final evaluation. Each search-selected architecture is reinitialized and trained under the same final Adam/early-stopping protocol, thereby removing inherited training age from the reported test comparison.

### Reviewer 4

**1. Negative transfer and local trapping.** We now state explicitly that tensor-shape compatibility is only a loading-safety condition; it provides no theoretical guarantee of positive transfer, global optimality, or finite-horizon Q-learning convergence. A validation gate compares inherited and freshly initialized compatible candidates under identical three-epoch budgets and propagates only the lower-validation-MSE branch. We present this as an empirical rollback safeguard, not a theorem.

**2. Decoupling controller, optimizer, reuse, and training age.** We added five matched arms: Q+Adam+gated reuse, Random+Adam+gated reuse, Q+SGD+gated reuse, Q+Adam+cold start, and Q+Adam+direct reuse. Data partitions, seeds, 12-episode horizon, candidate budget, and reward are fixed. Common from-scratch final retraining isolates architecture selection from optimizer-state inheritance and effective training age.

**3. Scalability and overhead.** The present table has 18 states and 7 actions (126 entries). For window length  $w$ , input dimension  $d$ , width  $H$ , and depth  $L$ , recurrent computation is approximately  $O(wL(H^2+dH))$ ; multi-lead inputs increase the first-layer term linearly, while an additional discrete hyperparameter with  $m$  levels multiplies the tabular state count by  $m$ . Mean updates and wall times are now reported; gated Q uses 4,128 mean updates versus 3,320 for cold-start Q.

**4. ECG5000 source.** We added the UCR/UEA archive citation, the ECG5000 Zenodo record (DOI 10.5281/zenodo.11186692), and PhysioNet provenance. The manuscript identifies BIDMC CHF record *chf07* and limits interpretation because beats from one source record are not independent patients.

**5. Recent methods.** Related work now discusses representative 2020–2026 forecasting architectures, including Informer, DLinear, PatchTST, TimesNet, and ModernTCN. We clarify that these methods are contextual comparators rather than claimed endpoint-equivalent ECG baselines unless rerun under the same split and objective.

### Reviewer 8

**Controlled experiments and seeds.** All five ablations were repeated for seeds {11, 22, 33, 44, 55}. We report mean results with 95% Student- $t$  confidence intervals and retain the official 4,500-beat test partition for one post-selection evaluation per arm. The experiment separates Q versus random selection, Adam versus SGD, and gated/direct/cold-start reuse.

**Key corrected result.** Q+Adam+gated reuse obtained test MSE  $0.02355 \pm 0.00162$ , MAE  $0.08920 \pm 0.00332$ , and  $R^2$   $0.96161 \pm 0.00265$  (95% CI half-widths). Cold-start Q was numerically best (MSE  $0.02288 \pm 0.00092$ ), but intervals overlap. We therefore removed the earlier causal/superiority claim: within the tested budget, the evidence does not establish a statistically resolved advantage for Q-learning or checkpoint inheritance.

**Editorial and reproducibility corrections.** The abstract, contributions, method, algorithm, experimental protocol, results, limitations, and conclusion were aligned with the controlled experiment. Seed lists, partition policy, stopping criteria, candidate/final budgets, compute counts, provenance, and uncertainty are now explicit. Claims are restricted to forecasting methodology rather than clinical validity.

*All changes are incorporated in the revised manuscript; this response intentionally reports both positive and null findings.*